

Access and manage workspaces

This article explains what a workspace is and how the workspace experience differs for Researchers and Research Leads. It also describes Research Lead tasks such as creating and deleting workspaces, managing members, and allocating additional quota.

Overview of a workspace

What is a Quantum workspace?

A workspace in Azure Quantum is a collection of resources used to run quantum applications and manage quantum workloads. You need a workspace to submit quantum programs to quantum hardware.

The Quantum OS displays different workspace information based on your role: Researcher or Research Lead.

- Researchers can view a single workspace. The workspace page includes the following sections:
 - **Basic information** - Shows workspace details such as status, region, and subscription ID.
 - **Members** - Shows the users who have access to the workspace.
 - **Quota allocations** - Shows the quota allocated to the workspace.
 - **Connections** - Provides an option for connecting the workspace to Visual Studio Code.
- Research Leads can view and manage all workspaces available to them. On a workspace details page, a Research Lead can view workspace information and perform workspace management tasks, such as managing members, allocating quota, and connecting to Visual Studio Code. For task-specific procedures, see [Manage workspaces and workspace settings as a Research Lead](#).

View workspaces in the Quantum OS

You can view a workspace in the Quantum OS based on your role. Researchers can view their assigned workspace, while Research Leads can view the workspaces available to them.

Prerequisites

- Access to sign in to the Quantum OS.

View a workspace

1. Sign in to the Quantum OS.
2. In the navigation pane, select **Workspace** if you're a Researcher, or **Workspaces** if you're a Research Lead.

What you see next depends on your role:

- If you're a Researcher, the Quantum OS opens your assigned workspace.
- If you're a Research Lead, the Quantum OS displays a list of available workspaces.

Manage workspaces and workspace settings as a Research Lead

As a Research Lead, you can perform the following tasks in the Quantum OS:

- **Manage workspaces**
 - Create a workspace
 - Delete a workspace
- **Manage members**
 - Add members to a workspace
 - Remove members from a workspace
- **Manage additional quota**
 - Allocate additional quota
- **Configure audit logging**
 - Link a Log Analytics workspace
- **Connect to a workspace in Visual Studio Code**
 - View a workspace in Visual Studio Code
 - Connect to a workspace by using a connection string

Create a workspace

1. Sign in to the Quantum OS.
2. In the navigation pane, select **Workspaces**.
3. Select **+ Create workspace**.
4. In the **Create workspace** dialog, enter a **Name** for the workspace.
5. Select an **Organization**.
6. Select **Next**.
7. For each quota type, enter a value in hours for:
 - **Standard quota**
 - **High quota**
8. Select **Next**.
9. Confirm the workspace details, and then select **Create workspace**.

Delete a workspace

1. Sign in to the Quantum OS.
2. In the navigation pane, select **Workspaces**.
3. From the **Workspace list**, select the checkbox for the workspace that you want to delete.
4. Select **Delete**.
5. Confirm you want to permanently delete the selected workspace, and then select **Delete**.

Add members to a workspace

1. Sign in to the Quantum OS.
2. In the navigation pane, select **Workspaces**.
3. From the **Workspace list**, select the workspace that you want to manage.
4. Select the **Members** tab.
5. Select **+ Add member**.
6. In the **Add member** dialog, select users in your directory or invite external guests to collaborate with your organization.
7. Select **Next**.

8. (Optional) Customize the invitation message that will be sent to invited users. Leave the default or write your own.
9. Select **Next**.
10. Review the details, and then select **Add members**.

Remove members from a workspace

1. Sign in to the Quantum OS.
2. In the navigation pane, select **Workspaces**.
3. From the **Workspace list**, select the workspace that you want to manage.
4. Select the **Members** tab.
5. Select the check boxes for the members that you want to remove.
6. Select **Remove**.

Allocate additional quota

You can add quota hours for each target and priority level in a workspace.

1. Sign in to the Quantum OS.
2. In the navigation pane, select **Workspaces**.
3. Do one of the following:
 - Select the checkbox next to a workspace in the **Workspace list**, and then select **Allocate quota**.
 - Select a workspace name to open the workspace details page, and then select **Allocate quota**.
4. In the **Allocate quota** dialog, enter the number of additional hours to add for each target and priority level.

The dialog shows the current allocation, used quota, additional hours, and new total for each target and priority level.

5. Select **Confirm**.

Link a Log Analytics workspace

You can link an existing Log Analytics workspace to collect audit logs from an Azure Quantum workspace.

1. Sign in to the Quantum OS.
2. In the navigation pane, select **Workspaces**.
3. From the **Workspace list**, select the workspace that you want to configure.
4. On the **Overview** tab, in the **Log Analytics** section, select **Link workspace**.
5. In the **Link Log Analytics workspace** dialog, select an existing workspace from the **Log Analytics workspace** list.
6. Select **Save**.

After you link the workspace, audit logs from the Azure Quantum workspace are collected in the selected Log Analytics workspace.

View a workspace in Visual Studio Code

You can use Visual Studio Code (VS Code) to connect to a specific Azure Quantum workspace outside the Quantum OS.

1. Sign in to the Quantum OS.
2. In the navigation pane, select **Workspaces**.
3. From the **Workspace list**, select a workspace that you want to use.
4. In the **Connection string** section, select **View workspace in VS Code** for either the **Primary connection string** or **Secondary connection string**.
5. When prompted to allow the **Microsoft Quantum Development Kit (QDK)** extension to open the link, select **Open**.
6. In the VS Code prompt, confirm the workspace that you want to add, and then select **Add Workspace**.

The workspace appears in the **Explorer** pane, under **QUANTUM WORKSPACES**.

[!NOTE] Researchers can also do this task from the **Connections** section of their workspace.

Connect to a workspace by using a connection string

You can use a connection string from the Quantum OS to connect to a specific Azure Quantum workspace in VS Code.

1. Sign in to the Quantum OS.
2. From the Navigation pane, select **Workspaces**.
3. From the **Workspace list**, select a workspace that you want to use.
4. On the workspace details page, in the **Connection string** section, select the **Copy** icon for either the **Primary connection string** or **Secondary connection string**.
5. Open VS Code.
6. Open the **View** menu and select **Command Palette**.
7. Enter `QDK: Connect to an Azure Quantum workspace`, and then select the command.
8. Select **Connection string**.
9. Paste the connection string that you copied into the input box, and then select **Enter**.

The workspace appears in the **Explorer** pane, under **QUANTUM WORKSPACES**.

[!NOTE] For more information about connection strings and their corresponding primary and secondary keys, see [Manage your access keys and connection strings](#).